

```
# =====
# EDGE DETECTION IN LOW CONTRAST IMAGES
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# 1. Upload Image
# -----
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

if img is None:
    print("Error: Image could not be loaded.")
else:

    # -----
    # 2. Canny on Original Low-Contrast Image
    # -----
    edges_before = cv2.Canny(
        img,
        50,
        150
    )

    # -----
    # 3. Contrast Enhancement using CLAHE
    # -----
    clahe = cv2.createCLAHE(
        clipLimit=2.0,
        tileGridSize=(8, 8)
    )

    enhanced = clahe.apply(img)

    # -----
    # 4. Gaussian Filtering
    # -----
    filtered = cv2.GaussianBlur(
        enhanced,
        (5, 5),
        0
    )

    # -----
    # 5. Canny after Preprocessing
    # -----
    edges_after = cv2.Canny(
        filtered,
        50,
        150
    )

    # -----
    # 6. Display Results
    # -----
    plt.figure(figsize=(15, 8))
```

```
plt.subplot(2, 3, 1)
plt.imshow(img, cmap="gray")
plt.title("Original Low-Contrast Image")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(edges_before, cmap="gray")
plt.title("Canny Before Preprocessing")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.hist(img.ravel(), bins=256)
plt.title("Original Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 3, 4)
plt.imshow(enhanced, cmap="gray")
plt.title("After CLAHE Enhancement")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(filtered, cmap="gray")
plt.title("After Gaussian Filtering")
plt.axis("off")

plt.subplot(2, 3, 6)
plt.imshow(edges_after, cmap="gray")
plt.title("Canny After Preprocessing")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# 7. Display Improved Histogram
# -----
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.hist(img.ravel(), bins=256)
plt.title("Before Enhancement")

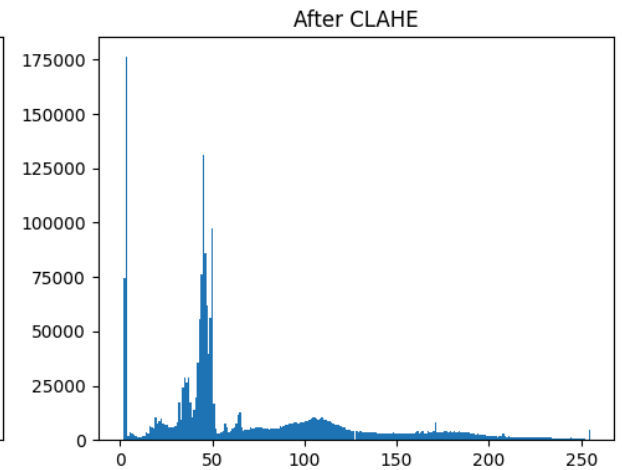
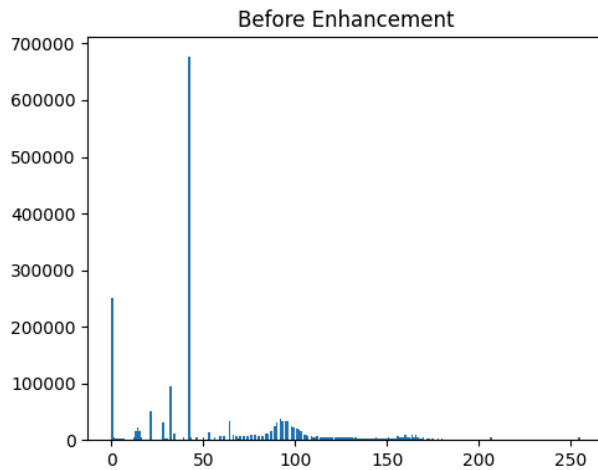
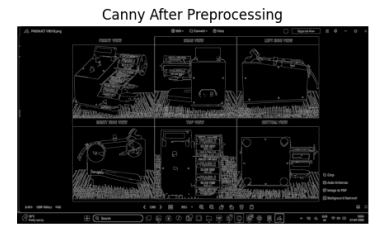
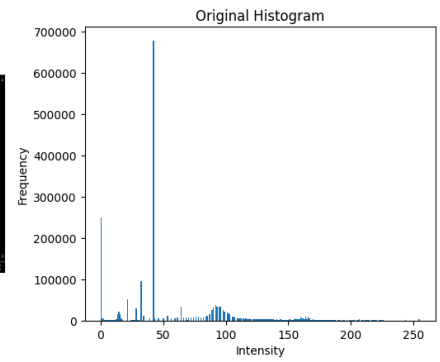
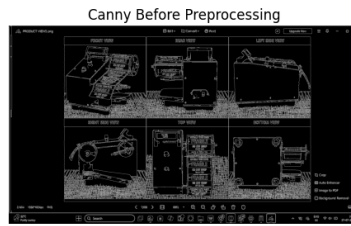
plt.subplot(1, 2, 2)
plt.hist(enhanced.ravel(), bins=256)
plt.title("After CLAHE")

plt.tight_layout()
plt.show()

print("Edge detection completed successfully.")
```

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Screenshot (152).png(image/png) - 1968062 bytes, last modified: 7/27/2026 - 100% done
Saving Screenshot (152).png to Screenshot (152).png



Edge detection completed successfully.